

Optimisation report — ludo_refine

Run

Generated 2026-09-10 16:30 — output folder: exports/optim/ludo_refine
Method: differential_evolution budget: 20 evaluations performed: 97 elapsed: 1 s backend: GPU
Decision variables: 24 wall distances: 50 cm eval grid: 30² over 250 cm, 300 rays/pixel

Key results

	initial	best #1	best #2
Score (lower is better)	0.2657	0.1411	0.2156
Uniformity U0 (u0)	73.4 %	85.9 %	78.4 %
FOV coverage	100 %	100 %	100 %
E_avg in FOV (mean over walls)	29,354 lx	12,794 lx	22,122 lx
Active LEDs	14	8	10
Drivers	14	8	10
Total current	42.0 A	24.0 A	30.0 A
E_min / E_max (last wall)	21,555 / 40,334 lx	10,989 / 16,071 lx	17,352 / 27,918 lx
ΔEV (stops)	0.90	0.55	0.69
penalty: coverage	0.0000	0.0000	0.0000

Best design improves the score by 46.9 % over the initial configuration (0.2657 → 0.1411).

The 'initial' column is the optimiser's starting point x_0 : the loaded scene when an existing design is refined, or the centre of the search box for a generated layout. The two best designs differ in at least one variable by more than 2 % of its range.

Problem definition

Scene / evaluation settings

Main camera: position (10, -4) cm, pitch 0°, FOV 75° × 60°
Emission: default 168 lm/LED, focus factor 0
Driver model: 6 V, 180 lm/W → 1,080 lm/A, max 13 A per LED, 1 LED(s) per driver

Objective

metric = u0, E_min percentile = 2 %, coverage weight = 1

Constraints (active)

none

Decision variables

variable	min	max	int	initial	best #1	best #2
group0.dx	-2	2		0	1.707	-1.308
group0.dy	-2	2		0	-0.3313	0.3455
group0.dz	-2	2		0	-0.4565	-0.4146
group0.drot_x	-15	15		0	3.335	4.467
group0.drot_y	-15	15		0	4.924	14.42
group0.drot_z	-20	20		0	6.411	2.982
group0.tilt[0]	-30	30		0	-24.91	-26.04
group0.tilt[1]	-30	30		0	4.914	7.762
group0.tilt[2]	-30	30		0	14.16	9.475
group0.tilt[3]	-30	30		0	17.73	23.31
group0.tilt[4]	-30	30		0	5.312	7.889
group0.tilt[5]	-30	30		0	-22.17	-18.46
group0.tilt[6]	-30	30		0	-24.98	1.697
group0.tilt[7]	-30	30		0	-10.62	-1.655
group0.tilt[8]	-30	30		0	25.65	16.65
group0.on[0]	0	1	●	1	0	0
group0.on[1]	0	1	●	1	1	1
group0.on[2]	0	1	●	0	0	1
group0.on[3]	0	1	●	1	1	1
group0.on[4]	0	1	●	1	0	1
group0.on[5]	0	1	●	1	1	0
group0.on[6]	0	1	●	0	0	1
group0.on[7]	0	1	●	1	1	0
group0.on[8]	0	1	●	1	0	0

Method & formulas

How a design is scored

Each candidate decision vector x is decoded into a regular saved configuration (same JSON schema as configs/*.json), the scene is built with the exact same code as the UI, and the wall illuminance E (lux) is ray-traced on a grid at every wall distance d in the list above. Only the cells inside the main-camera footprint (the FOV trapezoid, drawn in white on the images) are kept; cells with $E = 0$ are ignored for the uniformity metrics but counted for coverage.

Uniformity metrics over the lit FOV cells:

$$E_{min} = P_2(E), \quad E_{avg} = \bar{E}, \quad E_{max} = \max(E)$$

$$U_0 = \frac{E_{min}}{E_{avg}}, \quad U_1 = \frac{E_{min}}{E_{max}}, \quad CV = \frac{\sigma_E}{E_{avg}}$$

E_{min} uses the 2-th percentile of lit cells so a single Monte-Carlo-starved cell cannot dominate the metric.

Score (minimised):

$$S(x) = \frac{1}{N_d} \sum_d [1 - U_0]_d + 1 \left(1 - \min_d \text{cov}_d \right) + \sum_k P_k(x)$$

cov_d is the fraction of FOV cells receiving any light at distance d . Every constraint is a soft penalty $P_k \geq 0$ added to the score (no hard infeasibility), which keeps the landscape usable for population-based global search.

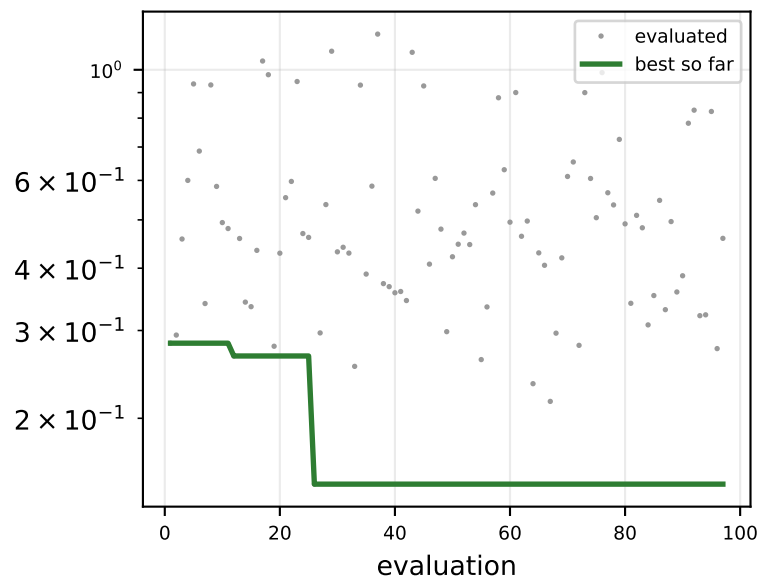
Active penalty terms:

Optimiser:

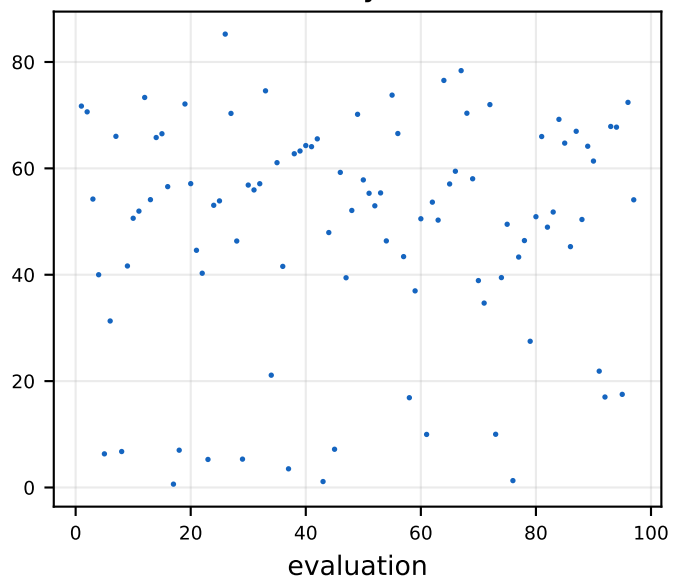
Differential evolution (scipy) samples a population inside the bounds, recombines candidates and keeps improvements; integer variables (on/off states, row/column counts) are rounded before decoding. Nelder-Mead and random search are local/simple alternatives used for refinement. The best design is exported after every improvement, so a stopped run is never lost.

Convergence

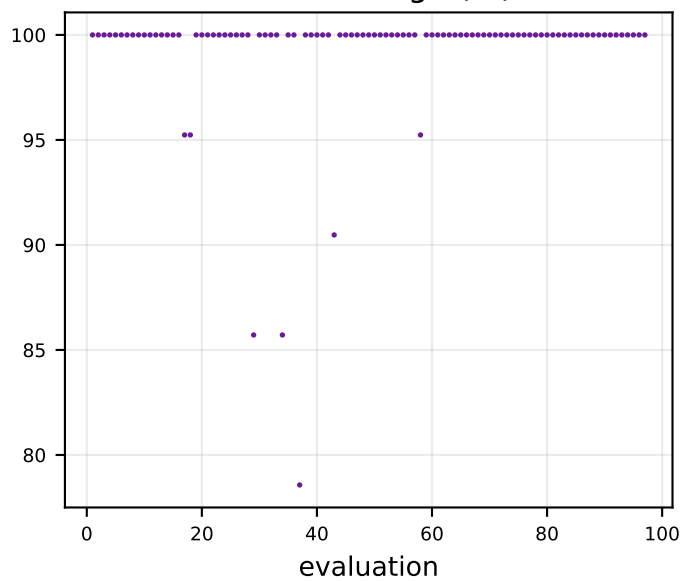
Score



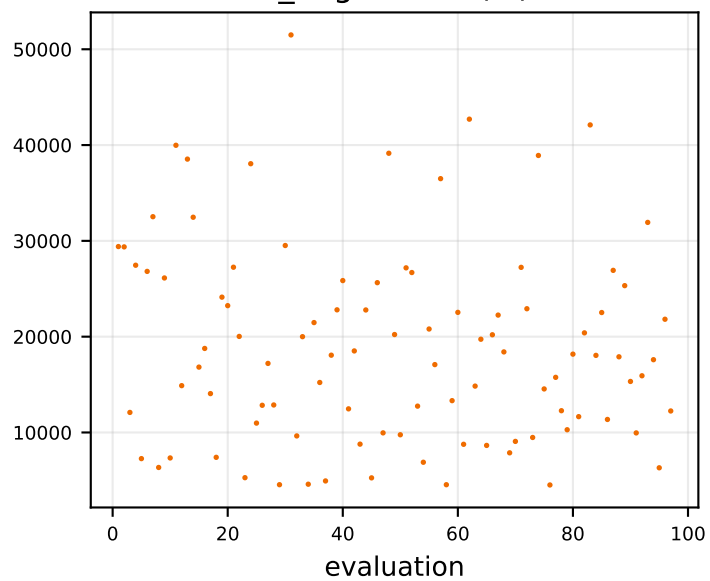
Uniformity U0 (%)



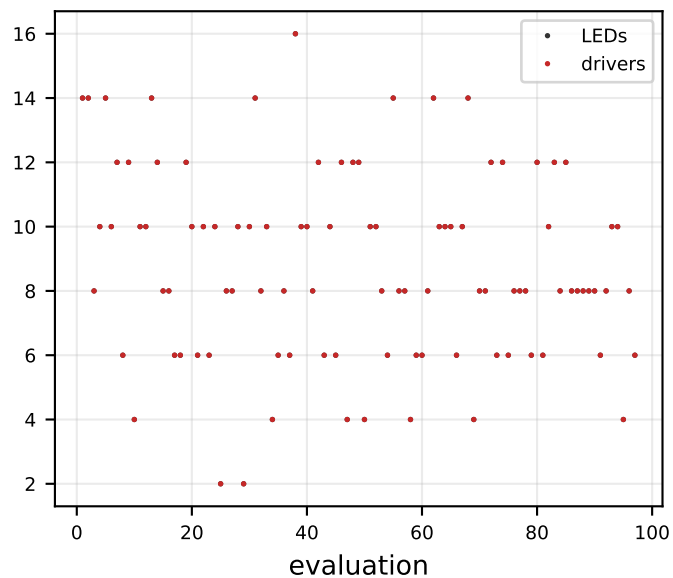
FOV coverage (%)



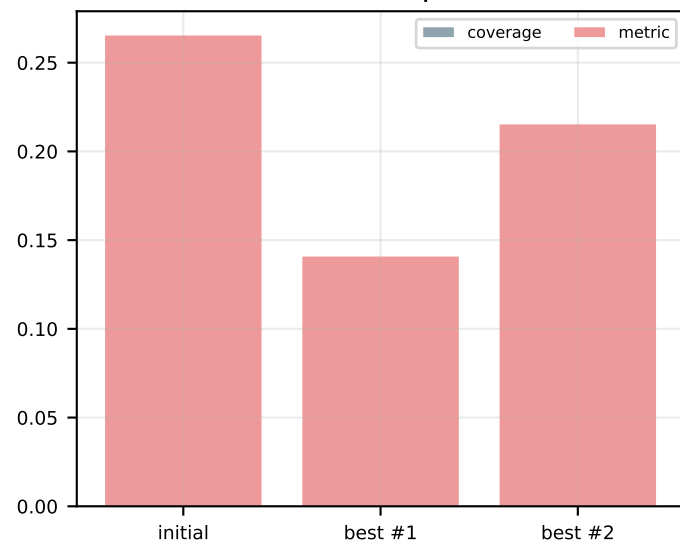
E_avg in FOV (lx)



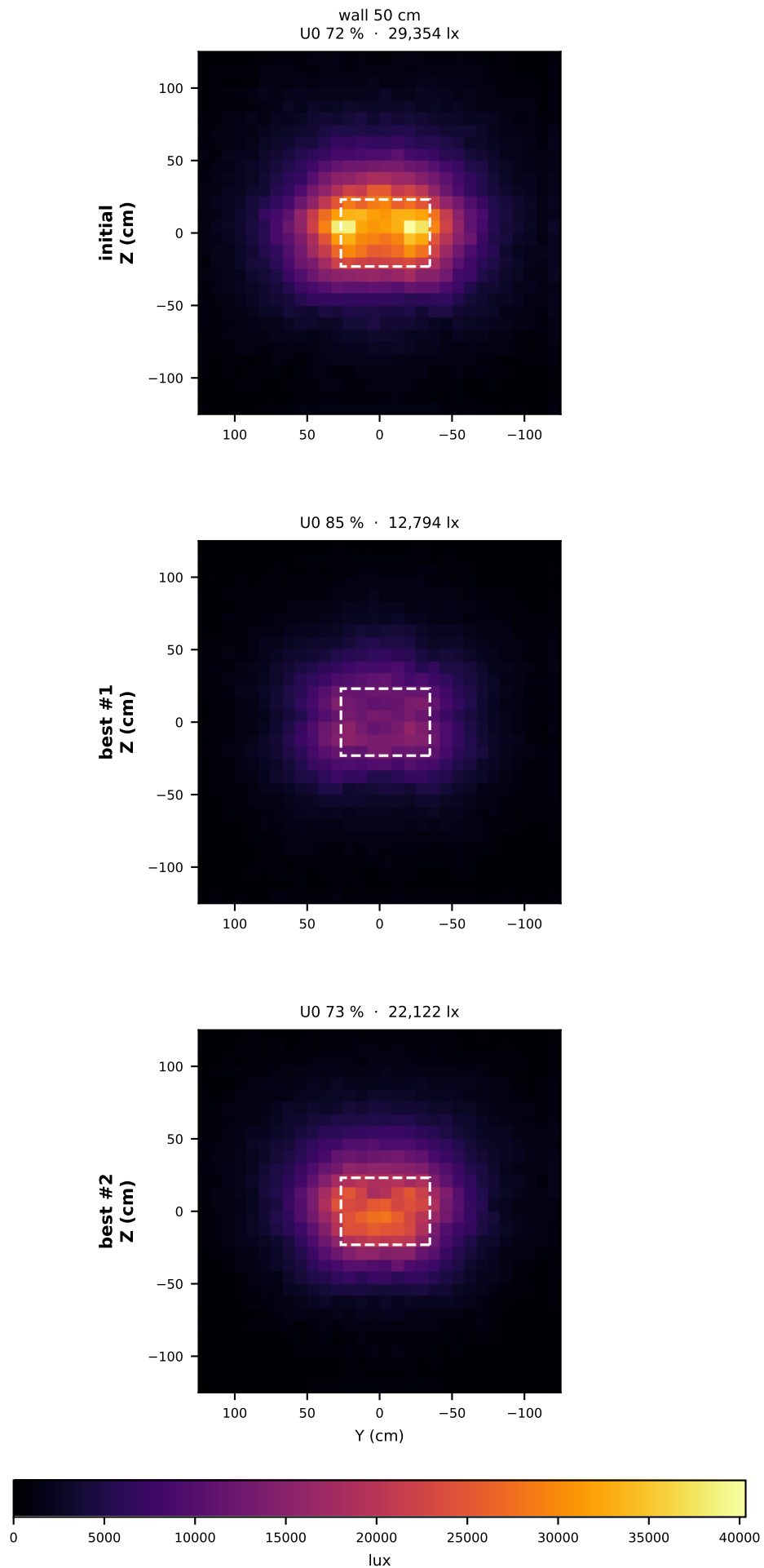
Active LEDs / drivers



Score decomposition

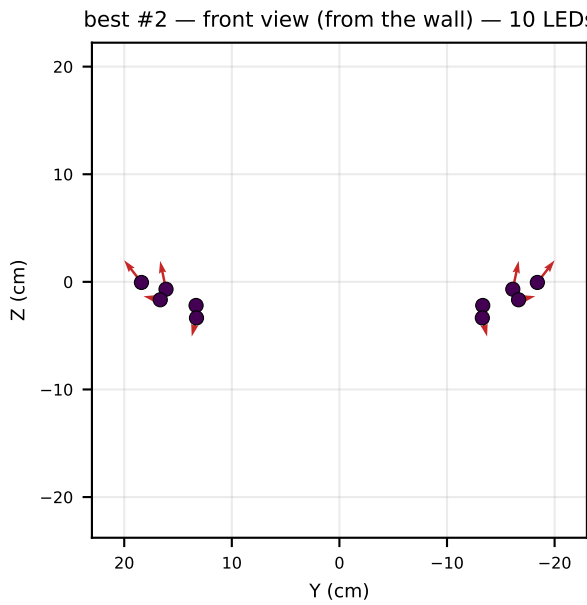
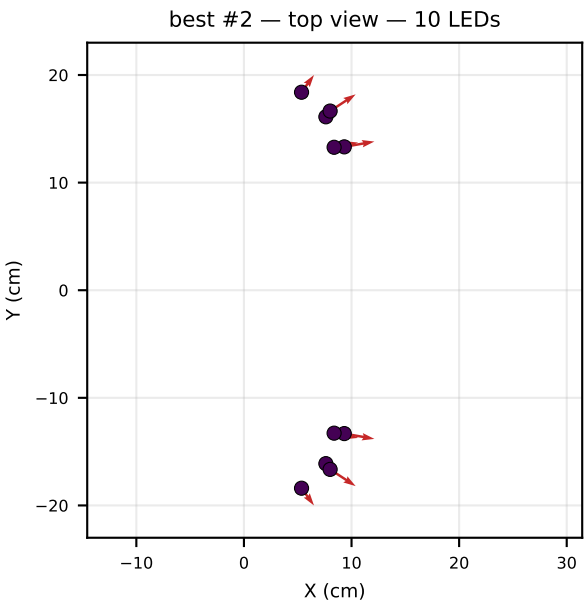
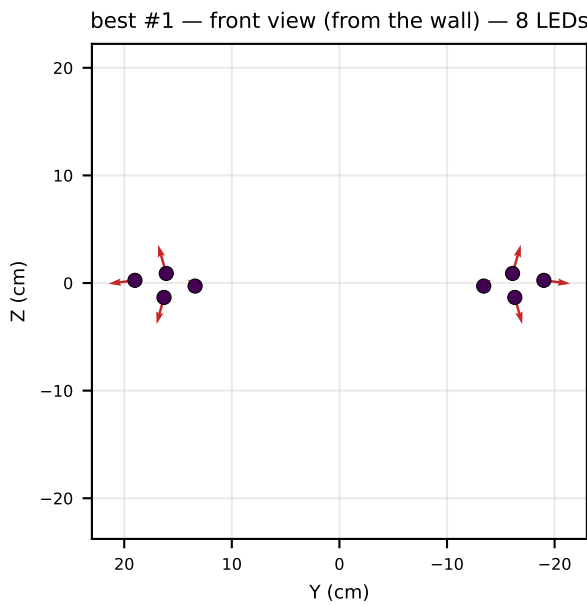
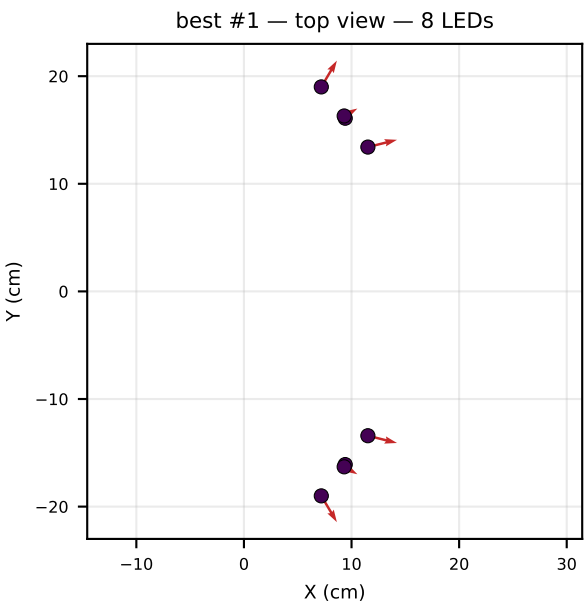
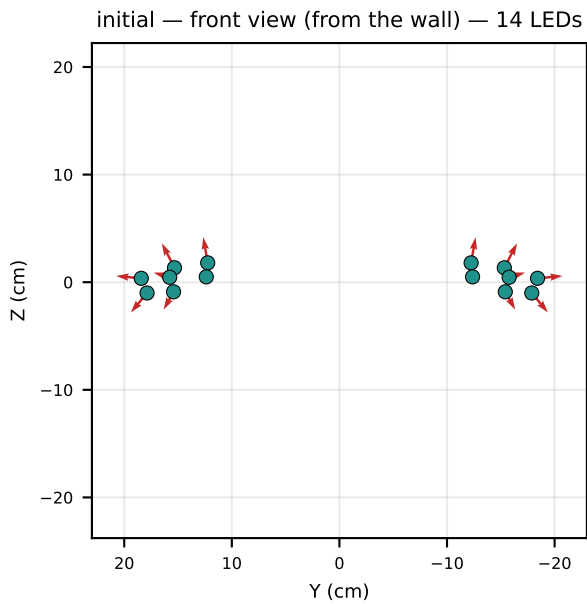
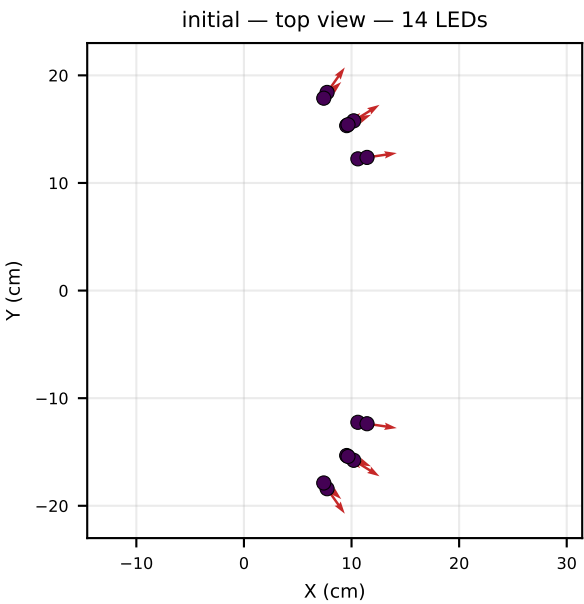


Wall illuminance — initial vs. best designs

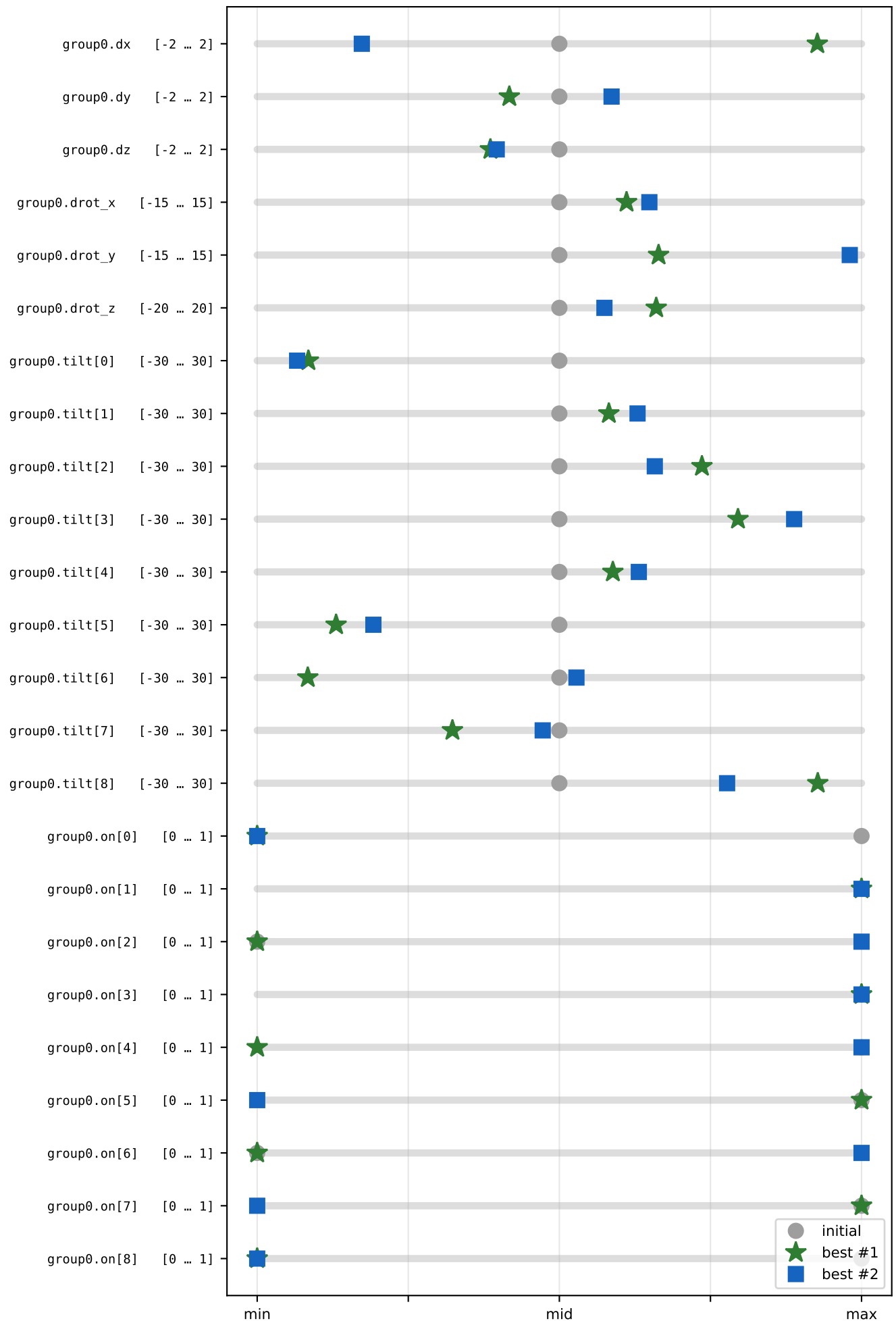


Dashed white: main-camera FOV footprint (metrics are computed inside it). Cyan: VIO camera footprint. Colour scale is shared per column.

LED layouts (active LEDs, arrows = beam axis)

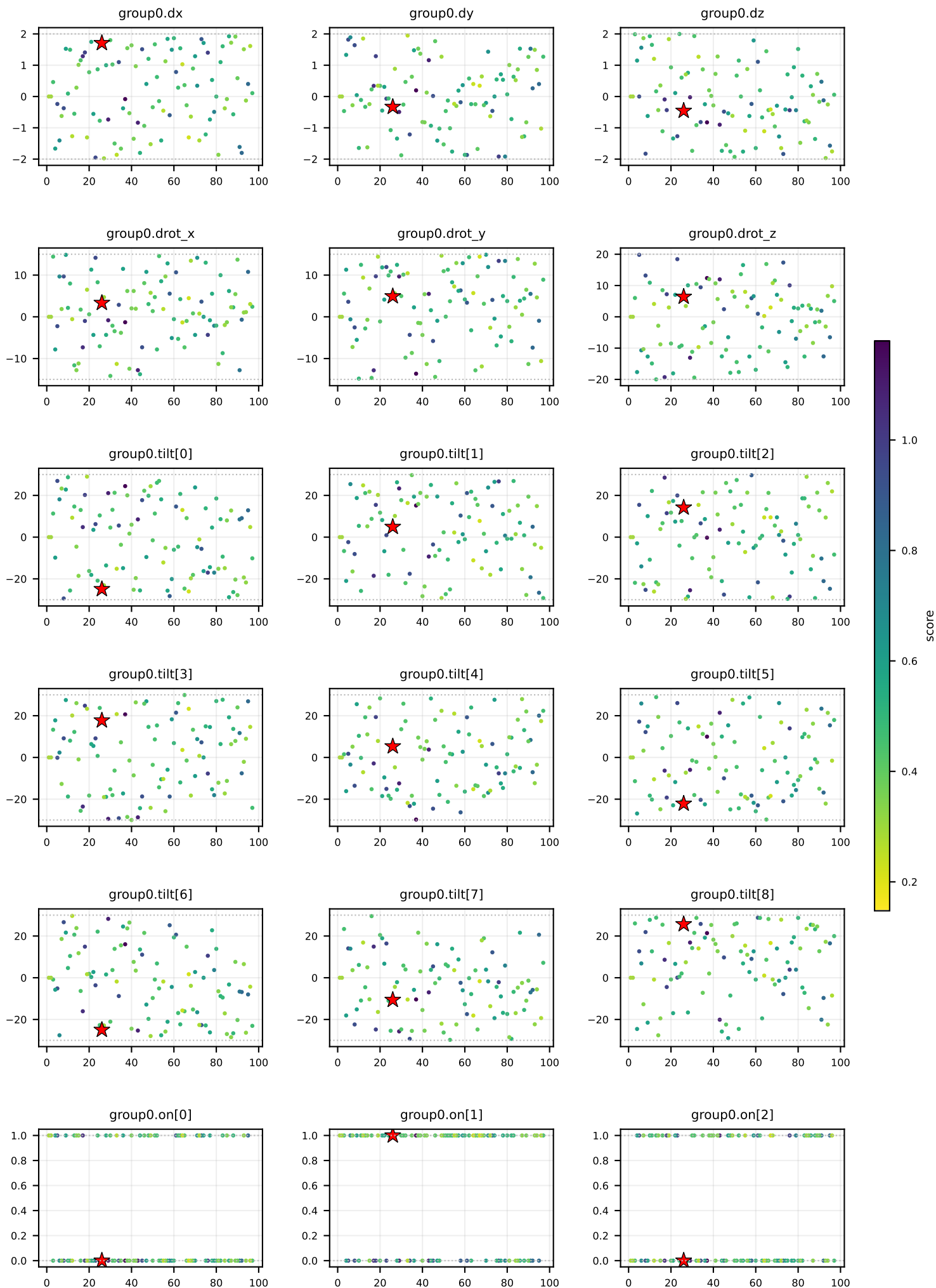


Optimised variables — position inside the search bounds

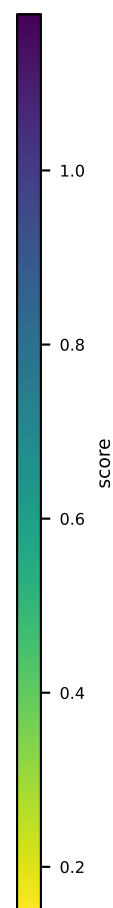
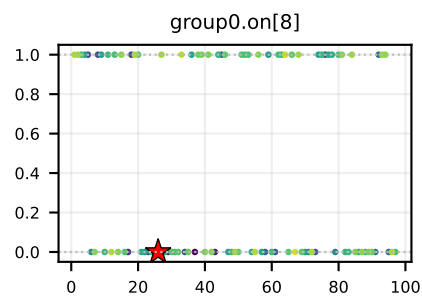
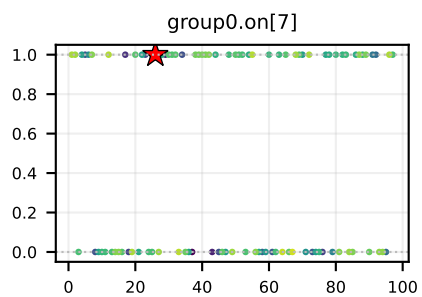
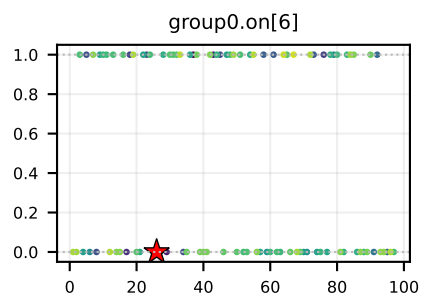
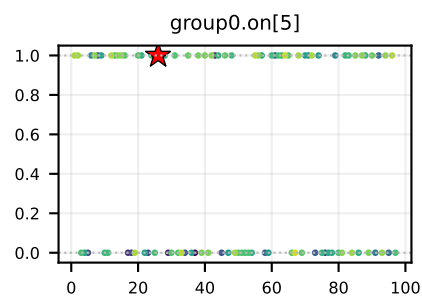
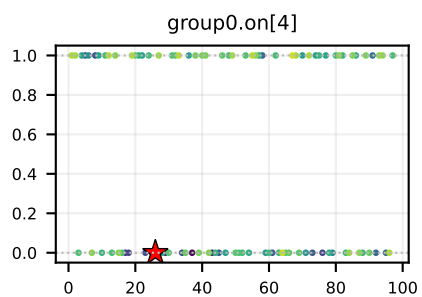
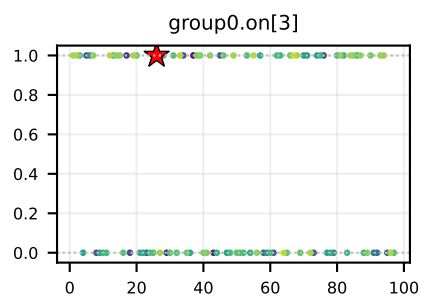


Values sitting on a bound suggest the search box may be too tight for that variable.

Optimised variables — explored values (colour = score, ★ = best)



Optimised variables — explored values (colour = score, ★ = best)



Files

best_config.json — best design, loadable in the UI (Project → Load) or via --evaluate
best2_config.json / initial_config.json — the other two designs compared in this report
history.csv — every logged evaluation with score, metrics and all variable values
summary.json — run metadata, best evaluation and best decision vector

initial: score=0.2657 U=73.4% cov=100% Eavg=29354lx LEDs=14 drv=14 I=42.0A

best #1: score=0.1411 U=85.9% cov=100% Eavg=12794lx LEDs=8 drv=8 I=24.0A

best #2: score=0.2156 U=78.4% cov=100% Eavg=22122lx LEDs=10 drv=10 I=30.0A